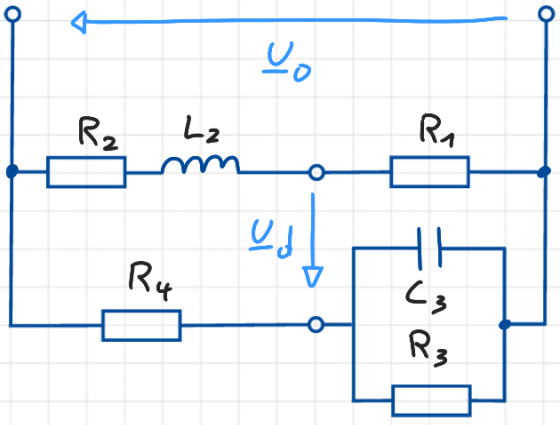
Messbrücke nach Maxwell-Wien



$$Z_1 = R_1$$
$$Z_2 = R_2 + \omega \cdot j \cdot L_2$$
$$Z_3 = \frac{1}{\frac{1}{R_3} + \omega \cdot j \cdot C_3}$$
$$Z_4 = R_4$$

$$\frac{Z_1}{Z_2} = \frac{Z_3}{Z_4} \longrightarrow$$

$$Re(Z_1 \cdot Z_4) = Re(Z_2 \cdot Z_3)$$
$$Im(Z_1 \cdot Z_4) = Im(Z_2 \cdot Z_3)$$

$$\frac{R_1}{R_2 + \omega \cdot j \cdot L_2} = \frac{\frac{1}{\frac{1}{R_3} + \omega \cdot j \cdot C_3}}{R_4}$$

$$\frac{R_1}{R_2 + \omega \cdot j \cdot L_2} = \frac{1}{R_4 \cdot \left(\frac{1}{R_3} + \omega \cdot j \cdot C_3\right)}$$

$$\frac{R_1}{R_2 + \omega \cdot j \cdot L_2} = \frac{1}{\frac{1}{R_3} R_4 + \omega \cdot j \cdot C_3 \cdot R_4}$$

$$\frac{R_1}{R_2 + \omega \cdot j \cdot L_2} = \frac{1}{\frac{R_4 + \omega \cdot j \cdot C_3 \cdot R_3 \cdot R_4}{R_3}}$$

$$\frac{R_1}{R_2 + \omega \cdot j \cdot L_2} = \frac{R_3}{R_4 + \omega \cdot j \cdot C_3 \cdot R_3 \cdot R_4}$$

$$R_1 \cdot (R_4 + \omega \cdot j \cdot C_3 R_3 R_4) = R_3 \cdot (R_2 + \omega \cdot j \cdot L_2)$$

$$\underbrace{R_1 R_4}_{Re} + \underbrace{\omega j \cdot R_1 C_3 R_3 R_4}_{Im} = \underbrace{R_2 R_3}_{Re} + \underbrace{\omega j \cdot L_2 R_3}_{Im}$$

Re: $R_1 \cdot R_4 = R_2 \cdot R_3 \longrightarrow \boxed{R_2 = \frac{R_1 \cdot R_4}{R_3}}$

Im: $\cancel{\omega j} \cdot R_1 C_3 \cancel{R_3} R_4 = \cancel{\omega j} \cdot L_2 \cdot \cancel{R_3} \longrightarrow \boxed{L_2 = R_1 \cdot C_3 \cdot R_4}$

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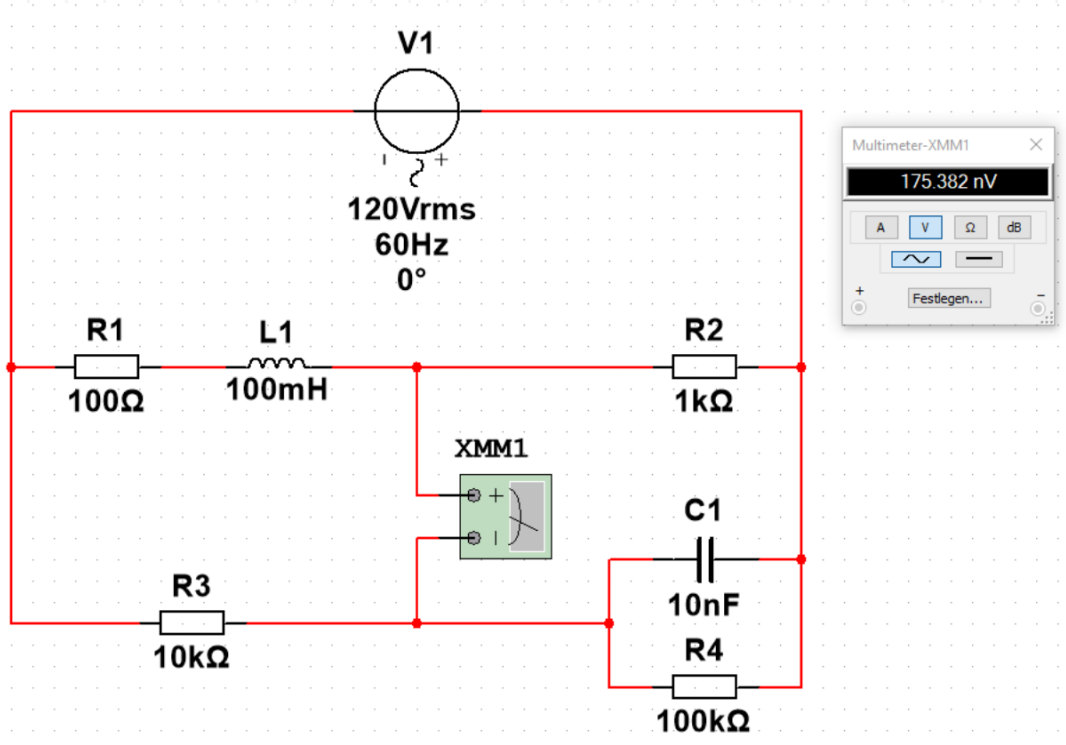
$R_1 = 1\,k\Omega \qquad R_3 = 100\,k\Omega \qquad C_3 = 10\,nF \qquad R_4 = 10\,k\Omega$

$$R_2 = \frac{R_1 \cdot R_4}{R_3} = \frac{1 \cdot 10^3 \cdot 10 \cdot 10^3}{100 \cdot 10^3}$$

$$R_2 = \underline{100\,\Omega}$$

$$L_2 = R_1 \cdot C_3 \cdot R_4 = 1 \cdot 10^3 \cdot 10 \cdot 10^{-9} \cdot 10 \cdot 10^3$$

$$L_2 = \underline{100\,mH}$$



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